One Shot Is All You Need. Or Is It?

Few-Shot SmolVLA Adaptation on LIBERO

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Abstract. I study how many robot demonstrations are needed to adapt SmolVLA to three held-out LIBERO tasks. A seen policy is first trained for 100k steps on all available `libero_90` data. Its zero-shot success is only **1/60**, but naive target fine-tuning with one demonstration reaches **109/120 (90.8%)**. This is close to the best result at ten demonstrations, **116/120 (96.7%)**. However, the same one-shot models retain only **37/180 (20.6%)** success on frozen seen-task probes, down from 80.0%. Target-LoRA uses 23.7× fewer trainable parameters but reaches 82.5% target success and 10.6% retention. Replay-LoRA falls to 55.8% and 1.1%, partly because target-only action statistics make replay actions badly scaled. Thus, one shot can be enough for the new task, but not for preserving old skills.

1 Question and main findings

Vision-language-action (VLA) models aim to connect images and language to robot actions. Large models can transfer across robots and tasks, but collecting a new teleoperation dataset remains costly [1,2]. I ask a simple question: *how far can a policy move the success–demonstration curve to the left?*

I use SmolVLA, a compact ∼450M-parameter VLA [2], and LIBERO, a benchmark for knowledge transfer in robot manipulation [3]. The fixed target tasks are the first three instructions in `libero_goal`. I report binary environment success, not a learned proxy.

My main findings are:

- The main gain happens before five demonstrations. One demo already gives 90.8% pooled target success; ten demos give 96.7%.
- More target data does not protect old skills. Corrected seen retention is 20.6% at $N = 1$, 11.7% at $N = 2$, and 0% at $N \geq 5$.
- Low-rank updates reduce trainable parameters, but they do not improve the target–retention trade-off in my frozen $N = 1$ comparison.

Research answer. The target cost curve is almost solved by one demonstration in this setup. The open problem is no longer only sample efficiency; it is *stable* adaptation.

2 Experimental setup

Data and split. I use the video-backed LeRobot v3 conversion `nvidia/LIBERO_LeRobot_v3` [4,5]. The seen set is the full available `libero_90`: 3,921 episodes, 569,249 frames, and 73 merged instruction rows. The three targets are held out by *instruction text*, not by row number:

Short name	Exact target instruction
Drawer	open the middle drawer of the cabinet
Bowl	put the bowl on the stove
Wine	put the wine bottle on top of the cabinet

Seen training. Starting from `lerobot/smolvla_base`, I freeze the vision encoder and VLM backbone, and train the Action Expert plus state/action projections. AdamW uses learning rate 10^{-4}, effective batch 32, bf16, 1k-step warm-up, then cosine decay. At 100k steps (3.2M samples), the final loss is 0.2349. Checkpoint selection uses only three `libero_90` probes: 21/30 at 60k, 23/30 at 80k, and 24/30 at 100k. The frozen weights have SHA-256 prefix `2cd510a594a8`.

Target protocol. For each budget $N \in \{1, 2, 5, 10, 25\}$, I take the first N demonstrations, never a hand-picked subset. Each cell has train seeds 42 and 123, followed by 20 fixed rollouts (seeds 1000–1019). The pooled denominator is therefore 120. $N = 0$ has one frozen model and 60 rollouts. Evaluation runs for at most 300 environment steps and executes ten actions from each 50-step predicted chunk.

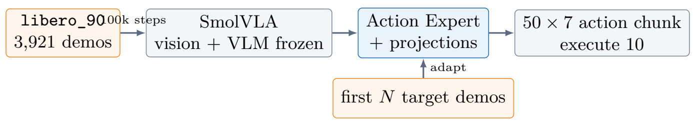


Figure 1. Training scope. Gray modules stay frozen; blue parameters are updated.

The gripper conversion is tested before training: $a_{\text{env}} = 1 - 2a_{\text{data}}$. A stored demonstration replays with 100% success, which catches the silent sign error.

[1] This project was prepared for the T-Lab 2026 VLA adaptation assignment (version 15 August 2026).

3 Cost curve and adaptation results

3.1 Preregistered prediction

Before final target evaluation, I committed the prediction that the largest gain would be from $N = 0$ to $N = 5$, with saturation from $N = 10$ to $N = 25$. I also predicted Replay-LoRA > Target-LoRA > naive fine-tuning at $N = 5$. The optimizer, LoRA rank, replay ratio, and checkpoint were fixed without target success.

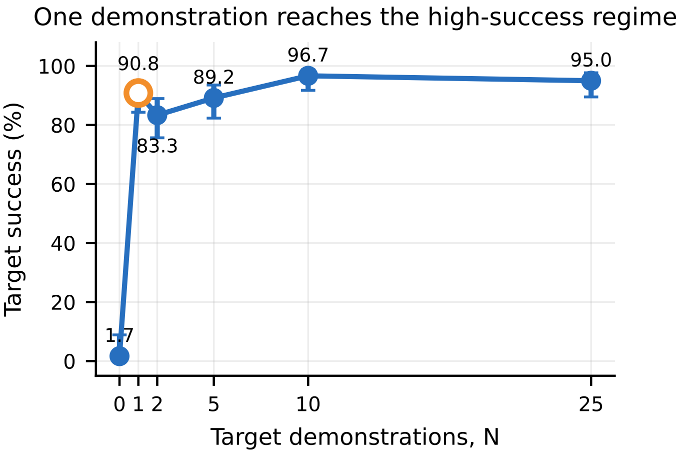


Figure 2. Pooled target success. Error bars are 95% Wilson intervals.

Naive baseline. The curve confirms the early-gain prediction, but the useful regime starts even earlier than expected. Success jumps from 1/60 zero-shot to 109/120 at $N = 1$. The $N = 1$ Wilson interval is [84.3, 94.8]%, so the point is already close to the later ceiling. $N = 2$ is worse than $N = 1$; extra demonstrations do not guarantee a monotonic result with only two train seeds.

N	Overall	Drawer	Bowl	Wine
0	1/60	1/20	0/20	0/20
1	109/120	36/40	40/40	33/40
2	100/120	30/40	38/40	32/40
5	107/120	31/40	40/40	36/40
10	116/120	39/40	40/40	37/40
25	114/120	40/40	40/40	34/40

Table 1. Exact target successes for naive adaptation.

The standard 0/5/10/25 curve would hide the most useful result: $N = 1$ gives a 89.2-point gain over zero-shot.

3.2 Corrected seen retention

A target-only model may look strong while forgetting its original tasks. I evaluate every adapted checkpoint on three frozen `libero_90` probes, ten seeds each, always with the original `libero_90` statistics.

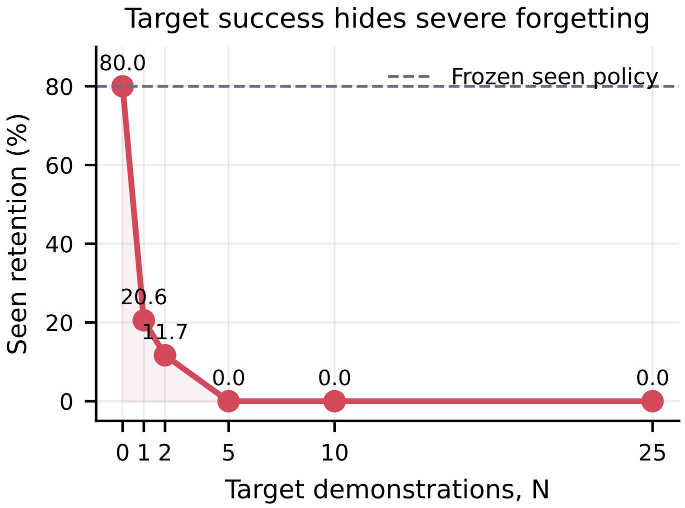


Figure 3. Corrected retention. $N = 0$ is the frozen 24/30 reference; each adapted point pools 180 rollouts.

Retention collapses as target exposure grows: 37/180 at $N = 1$, 21/180 at $N = 2$, and 0/180 for $N = 5, 10, 25$. Across the 30 adapted cells, target success and retention have Pearson $r = 0.026$. Target success alone is therefore a poor measure of a useful lifelong policy.

3.3 Task 2: two low-rank methods

Because the naive curve is near ceiling, I use the allowed $N = 1$ setting. **Target-LoRA** puts rank-64 adapters in the Action Expert and trains the projections [6]. **Replay-LoRA** adds a fixed 25% replay stream from `libero_90`; it never uses other target demos and replay is not selected by the retention probes. All 12 models start independently from the same frozen checkpoint.

Method	Target (%)	Retain (%)	Params trainable	Wall (s/cell)	VRAM (MiB)
Naive	**90.8**	**20.6**	99.9M	**162.8**	7,540
Target-LoRA	82.5	10.6	4.22M	244.0	**6,944**
Replay-LoRA	55.8	1.1	4.22M	309.3	**6,944**

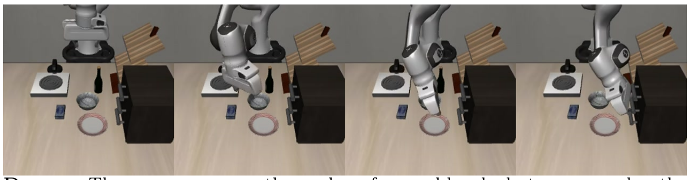
Table 2. Matched $N = 1$ result. Wall time includes concurrent load.

Neither method beats naive fine-tuning. Target-LoRA uses 23.7× fewer trainable parameters and 7.9% less reserved VRAM, but loses 8.3 target points and 10.0 retention points. Replay-LoRA loses 35.0 and 19.4 points. For Drawer, its final losses reach 1.44×10^{10} and 2.72×10^{10}: the one target demo has gripper standard deviation 10^{-6}, so applying target-only statistics to mixed seen replay creates extreme normalized actions. This explains the numerical failure, but not all of the retention loss on Bowl and Wine.

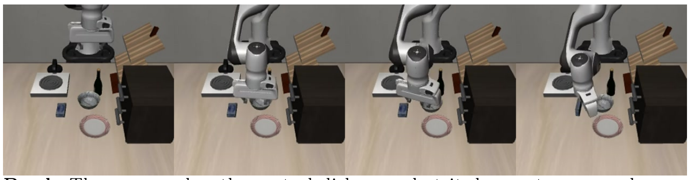
Parameter efficiency is not retention. A small update can still move the policy in a harmful direction, and replay is unsafe when data sources do not share a common action scale.

4 Three real failures

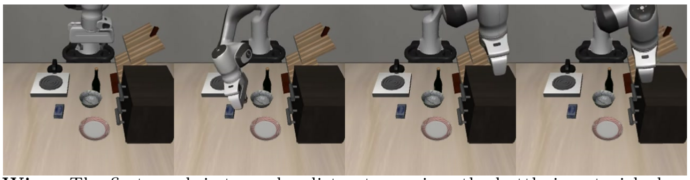
All strips below come from failed frozen-policy zero-shot rollouts at evaluation seed 1000. Frames are at 0, 5, 10, and 15 seconds; no successful run was selected.



Drawer. The arm moves over the work surface and bowls, but never reaches the middle drawer handle.



Bowl. The arm reaches the central dish area, but it does not grasp and move the bowl to the stove.



Wine. The first reach is toward a distractor region; the bottle is not picked or placed on the cabinet.

These videos suggest that zero-shot failure is not one single problem. The Drawer and Wine runs fail before contact, while Bowl reaches a plausible area but does not complete the grasp–place sequence.

4.1 Hypotheses and separating tests

1. Drawer: spatial grounding vs. low-level control. My hypothesis is that the language token 'middle drawer' is not grounded to the correct handle. I would compare the original image with a handle crop and a point-marked image, keeping the initial state and instruction fixed. The key metric is end-effector distance to the handle before any contact. A better approach under crops would support a perception/grounding failure; unchanged distance would point to the policy or action decoder.

2. Bowl: grasping vs. relational planning. The reach is plausible, so the failure may be the grasp rather than the instruction 'on the stove.' I would create two matched starts: the normal scene and the same scene with the bowl already in the gripper. Success only from the second start would isolate grasping. Failure in both starts would support a place-relation or long-horizon planning problem.

3. Wine: object identity vs. place horizon. The early motion appears aimed at a distractor. I would cross two interventions: swap the bottle/distractor colours, and start with the bottle already grasped. If the reach follows colour, object identity is the problem. If the grasped start still cannot place the bottle, the main bottleneck is the cabinet-top relation or action horizon.

4.2 Does the policy read language?

I replay the same 60 initial-state fingerprints with the correct target instruction and with an instruction from another task. Correct text gives 1/60 successes; wrong text gives 0/60. This is too sparse for a success-only claim, so I also compare actions. The mean paired action-sequence L_2 distance is 0.582 for Drawer, 0.998 for Bowl, and 0.966 for Wine. Thus, language changes behavior, even though zero-shot competence is weak.

Task	Correct	Wrong	Mean action L_2
Drawer	1/20	0/20	0.582
Bowl	0/20	0/20	0.998
Wine	0/20	0/20	0.966

Table 3. Paired language control on identical starts.

4.3 A normalization failure that looked like forgetting

An early retention run used target-overlay statistics and produced 0/900. A controlled check showed that even the frozen seen policy drops from 12/15 with seen statistics to 0/60 with target-overlay statistics. Therefore, 0/900 is a deployment-normalization failure, not evidence of weight forgetting. All retention numbers in this paper use the original `libero_90` statistics.

5 Predictions, rejected ideas, and lessons

5.1 Predictions vs. results

- **Correct:** I predicted the largest gain between zero and five demos, followed by saturation. The gain is 87.5 points at $0 \to 5$; $N = 10 \to 25$ changes by -1.7 points. Adding $N = 1$ shows that almost all of the gain arrives after the first demonstration.
- **Wrong at $N = 1$:** I expected Replay-LoRA > Target-LoRA > naive at low budget. The observed order is exactly the reverse. I underestimated two effects: source-dependent action normalization and how strongly small target updates can damage the seen policy.
- **Partly supported:** wrong language changes the trajectory and removes the only zero-shot success. Still, 1/60 vs. 0/60 is not enough to claim strong instruction understanding.

The preregistration predicted the method order at $N = 5$, which I did not run after the $N = 1$ ceiling experiment. I therefore report the $N = 1$ falsification as evidence, not as a formal $N = 5$ test.

5.2 Idea graveyard

Target-based hyperparameter sweeps. A sweep could make LoRA look better, but it would tune on the three test tasks. I rejected it and kept rank 64, learning rate, steps, and 25% replay fixed.

Target-overlay statistics for every deployment. This looked simple, but the 2-by-2 control showed that statistics alone can zero a frozen policy. I rejected the earlier retention result and reran the protocol with `libero_90` statistics.

Full VLM fine-tuning. I considered updating the whole 450M model, but rejected it because the Action Expert baseline was already near ceiling, while full updates would add compute and a larger forgetting risk without testing the key low-data idea.

5.3 Most surprising result

The surprising result is not only that one demo works. It is that *high target success and severe forgetting happen together*. At $N = 1$, the adapted policy gains 89.2 target points but loses 59.4 seen points. This suggests that the pretrained representation is useful, while the action distribution is easy to overwrite.

6 Scope, limitations, and conclusion

6.1 Bonus status

I selected reward-free evaluation (Bonus B) as the next extension, but did not finish a learned video critic or the Robometer comparison. I make no bonus claim. The next honest test is to rank the frozen and adapted checkpoints without environment rewards, then compare this order with true success.

6.2 Limitations

There are only three target tasks and two training seeds. Twenty rollouts per task and cell still give wide intervals, and three seen probes are sentinels, not a full `libero_90` evaluation. The $N = 1$ LoRA comparison tests parameter efficiency, not a complete search for the best adaptation method. Replay-LoRA's normalization bug is diagnosed but not repaired in these final numbers. Finally, all experiments are in simulation; real-robot noise may change the one-shot result.

6.3 Conclusion

I trained a reproducible SmolVLA seen expert and measured the complete naive cost curve, including the extra $N = 1, 2$ ceiling regime. One target demonstration is enough for 90.8% pooled success, but it is not enough for a stable lifelong policy. Neither Target-LoRA nor naive seen replay fixes this trade-off under the frozen protocol. The next method should use a common, frozen action coordinate system and an explicit constraint to preserve the seen policy. The central lesson is simple: **one shot is enough to learn the target, but not enough to remember everything else.**

6.4 Reproducibility

The first-page links provide the complete code, exact commands, frozen predictions, manifests, checksums, model weights, and result artifacts. The repository test suite passes 296 tests (8 skipped). The technical appendix contains per-task/per-seed results, training curves, hashes, and the full normalization audit.

References

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